

Practical No.13

MOHAMMAD QASIM

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Objective:

Statistical Tests in Python

Dataset: Impact of Teaching Method and Gender on Student Performance then find.

1. **Q: Use Shapiro-Wilk test to check the normality of Score_Before .**

H_0 : Score is normally distributed.

2. **Q: Examine whether the variances of Score are equal across the two teaching methods (Online and Offline).**

Perform both Bartlett's and Levene's tests. State your null and alternative hypotheses and interpret the p-values.

3. **Q: Perform t-tests for the following:**

A) Perform a one-sample t-test to check whether the sample mean of Score is significantly different from 70. Interpret your result.

B) Do the average scores differ between students taught Online and Offline? Perform an independent two-sample t-test and interpret the result.

C) Perform a paired t-test to determine if there is a significant improvement in scores after training.

4. **Q: Compare pass rates between methods (proportion test)**

Compare the proportion of students who passed (Passed = 1) between the two teaching methods (Online and Offline). Perform a two-sample z-test for proportions and interpret the findings.

5. **Q: Chi-square test of independence**

Is there an association between Gender and Teaching Method? Perform a chi-square test of independence using a contingency table of Gender $\times$ Method. State the hypotheses and interpret the result.

6. **Q: Test whether the teaching method has a significant effect on student Score.**

Perform a one-way ANOVA and interpret the ANOVA table. Then apply Tukey's HSD post-hoc test to identify pairwise differences.

7. **Q: Two-way ANOVA for effects of Method , Gender , and their interaction on Score .**

Examine whether both Method and Gender, and their interaction, have significant effects on Score. Perform a two-way ANOVA and interpret the main and interaction effects.

8. **Q: Non-parametric alternative of Two sample independent t test (Mann-Whitney U test)**

When the assumption of normality is not met, perform a Mann-Whitney U test to compare the median Score between the Online and Offline groups. Interpret the U-statistic and p-value.

9. **Q: Non-parametric alternative of Paired t-test (Wilcoxon signed-rank test)**

As a non-parametric alternative to the paired t-test, perform a Wilcoxon signed-rank test to compare Score and After_Training. Interpret the result.

Enter the given data

```
In [5]: import pandas as pd
import numpy as np

df = pd.read_csv('Statistical_Tests_dataset.csv')
df.head(8)
```

```
Out[5]:
```

	Gender	Method	Score	Score_After_Training	Pass
0	Female	Online	28	29	0
1	Male	Online	55	59	1
2	Male	Online	60	63	1
3	Female	Online	37	40	0
4	Female	Online	52	54	1
5	Male	Online	52	55	1
6	Female	Online	33	36	0
7	Female	Online	43	44	0

```
In [6]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99 entries, 0 to 98
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Gender                99 non-null    object  
1   Method                99 non-null    object  
2   Score                 99 non-null    int64   
3   Score_After_Training  99 non-null    int64   
4   Pass                 99 non-null    int64   
dtypes: int64(3), object(2)
memory usage: 4.0+ KB
```

Q: Use Shapiro-Wilk test to check the normality of Score_Before .

\$H_0\$: Score is normally distributed.

```
In [8]: from scipy import stats
stat, p = stats.shapiro(df['Score'])
print(f"Shapiro-Wilk W={stat:.4f}, p={p:.4f}")
if p>0.05:
    print('Conclusion: Fail to reject H0 – data is approximately normal.')
else:
    print('Conclusion: Reject H0 – data not normal.')
```

Shapiro-Wilk W=0.9452, p=0.0004
Conclusion: Reject H0 – data not normal.

Q: Examine whether the variances of Score are equal across the two teaching methods (Online and Offline).

Perform both Bartlett's and Levene's tests. State your null and alternative hypotheses and interpret the p-values.

```
In [9]: online = df[df['Method']=='Online']['Score']
offline = df[df['Method']=='Offline']['Score']
b_stat, b_p = stats.bartlett(online, offline)
l_stat, l_p = stats.levene(online, offline)
print('Bartlett:', "Statistic: ", b_stat, ", P-value: ", b_p)
print('Levene: ', "Statistic: ", l_stat, ", P-value: ", l_p)
```

Bartlett: Statistic: 1.734764380273997 , P-value: 0.18780358404781328
Levene: Statistic: 0.14604634236763836 , P-value: 0.7035521342956923

Q: Perform t-tests for the following:

- A) Perform a one-sample t-test to check whether the sample mean of Score is significantly different from 70. Interpret your result.
- B) Do the average scores differ between students taught Online and Offline? Perform an independent two-sample t-test and interpret the result.
- C) Perform a paired t-test to determine if there is a significant improvement in scores after training.

```
In [10]: # One-sample
t1, p1 = stats.ttest_1samp(df['Score'], 70)
print('One-sample t-test: t, p =', t1, p1)
# Two-sample independent
t2, p2 = stats.ttest_ind(online, offline, equal_var=True)
print('Two-sample t-test: t, p =', t2, p2)
# Paired
t3, p3 = stats.ttest_rel(df['Score'], df['Score_After_Training'])
print('Paired t-test: t, p =', t3, p3)
```

One-sample t-test: t, p = -10.419089789548275 1.4933079193064338e-17
Two-sample t-test: t, p = -7.678780457563491 9.130903077650608e-11
Paired t-test: t, p = -12.582015672617471 3.5318742157732294e-22

Q: Compare pass rates between methods (proportion test)

Compare the proportion of students who passed (Passed = 1) between the two teaching methods (Online and Offline). Perform a two-sample z-test for proportions and interpret the findings.

```
In [14]: from statsmodels.stats.proportion import proportions_ztest
count = np.array([df[df['Method']=='Online']['Pass'].sum(), df[df['Method']=='Offline']['Pass'].sum()])
nobs = np.array([len(df[df['Method']=='Online']), len(df[df['Method']=='Offline'])])
stat, pval = proportions_ztest(count, nobs)
print('Counts:', count, 'N:', nobs)
print('Proportion z-test: z, p =', stat, pval)
```

Counts: [24 35] N: [34 35]
Proportion z-test: z, p = -3.4697094042513337 0.0005210217431105547

Q: Chi-square test of independence

Is there an association between Gender and Teaching Method? Perform a chi-square test of independence using a contingency table of Gender × Method. State the hypotheses and interpret the result.

```
In [15]: table = pd.crosstab(df['Gender'], df['Method'])
from scipy import stats
chi2, p, dof, exp = stats.chi2_contingency(table)
print('Table:\n', table)
print('Chi2=', chi2)
print('p-value=', p)
print('dof =', dof)
print('Expected frequencies:\n', exp)
```

Table:

Method	Offline	Online	Self
Female	20	15	14
Male	15	19	16

Chi2= 1.308239753044075
p-value= 0.5198994369852212
dof = 2
Expected frequencies:
[[17.32323232 16.82828283 14.84848485]
[17.67676768 17.17171717 15.15151515]]

Q: Test whether the teaching method has a significant effect on student Score.

Perform a one-way ANOVA and interpret the ANOVA table. Then apply Tukey's HSD post-hoc test to identify pairwise differences.

```
In [16]: import statsmodels.api as sm
from statsmodels.formula.api import ols
```

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd
model = ols('Score ~ Method', data=df).fit()
print(sm.stats.anova_lm(model, typ=2))
```

	sum_sq	df	F	PR(>F)
Method	14828.946507	2.0	89.703734	1.071976e-22
Residual	7934.891877	96.0	NaN	NaN

```
In [17]: tukey = pairwise_tukeyhsd(df['Score'], df['Method'])
print(tukey)
```

```
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
Offline Online -19.6555 0.0 -24.8671 -14.4438 True
Offline Self -29.4476 0.0 -34.8326 -24.0626 True
Online Self -9.7922 0.0001 -15.2136 -4.3707 True
=====
```

Q: Two-way ANOVA for effects of Method, Gender, and their interaction on Score.

Examine whether both Method and Gender, and their interaction, have significant effects on Score. Perform a two-way ANOVA and interpret the main and interaction effects.

```
In [18]: model2 = ols('Score ~ C(Method) + C(Gender) + C(Method):C(Gender)', data=df).fit()
print(sm.stats.anova_lm(model2, typ=2))
```

	sum_sq	df	F	PR(>F)
C(Method)	15125.973451	2.0	94.225811	4.336930e-23
C(Gender)	330.962878	1.0	4.123403	4.514932e-02
C(Method):C(Gender)	139.331379	2.0	0.867952	4.231830e-01
Residual	7464.597619	93.0	NaN	NaN

Q: Non-parametric alternative of Two sample independent t test (Mann-Whitney U test)

When the assumption of normality is not met, perform a Mann-Whitney U test to compare the median Score between the Online and Offline groups. Interpret the U-statistic and p-value.

```
In [19]: u, up = stats.mannwhitneyu(online, offline, alternative='two-sided')
print('Mann-Whitney U, p =', u, up)
```

Mann-Whitney U, p = 112.5 7.135872523782862e-09

Q: Non-parametric alternative of Paired t-test (Wilcoxon signed-rank test)

As a non-parametric alternative to the paired t-test, perform a Wilcoxon signed-rank test to compare Score and After_Training. Interpret the result.

```
In [20]: w, wp = stats.wilcoxon(df['Score'], df['Score_After_Training'])
print('Wilcoxon W, p =', w, wp)
```

Wilcoxon W, p = 0.0 2.177458322500681e-15

Interpretation :

1. Use Shapiro-Wilk test to check the normality of Score_Before.

\$H_0\$: Score is normally distributed.

\$H_1\$: Score is **not** normally distributed.

Shapiro-Wilk W=0.9452, p=0.0004

Conclusion: Here p-value is less than 0.05 so we Reject \$H_0\$ and data is **not** normal.

2. Examine whether the variances of Score are equal across the two teaching methods (Online and Offline).

Performed both Bartlett's and Levene's tests.

$$H_0: \sigma_{\text{Online}}^2 = \sigma_{\text{Offline}}^2$$

v/s

$$H_1: \sigma_{\text{Online}}^2 \neq \sigma_{\text{Offline}}^2$$

Bartlett: Statistic: 1.734764380273997 , P-value: 0.18780358404781328

Levene: Statistic: 0.14604634236763836 , P-value: 0.7035521342956923

In both test Bartlett and Levene the p-value is greater than 0.05 so we accept the H_0 that means **variances of Score are equal** across the two teaching methods

3. Performed t-tests for the following:

$$H_0: \mu = \mu_0$$

v/s

$$H_1: \mu \neq \mu_0$$

One-sample t-test: $t = -10.419089789548275$, $p = 1.4933079193064338e-17$

A) For one-sample t-test the p-value is less than 0.05 so we reject H_0 and **sample mean of Score is significantly different from 70.**

$$H_0: \mu_x = \mu_y$$

v/s

$$H_1: \mu_x \neq \mu_y$$

Two-sample t-test: $t = -7.678780457563491$, $p = 9.130903077650608e-11$

B) Here p-value is less than 0.05 so we reject H_0 that the **average scores differ between students taught Online and Offline.**

$$H_0: \mu_{\text{before}} = \mu_{\text{after}}$$

v/s

$$H_1: \mu_{\text{before}} \neq \mu_{\text{after}}$$

Paired t-test: $t = -12.582015672617471$, $p = 3.5318742157732294e-22$

C) Here we perform paired t-test to determine the significant improvement in scores after training and p-value is less than 0.05 so we reject H_0 that we can say the students get the **significant improvement in scores after training.**

4. Compare pass rates between methods (proportion test)

Here the 24 students from 'online Method' are 'Passed' in out of 34 then_

$$p_1 = 24 / 34 = 0.706$$

while 35 students from 'Offline Method' are 'Passed' out of 35_

$$p_2 = 35 / 35 = 1$$

We state the Hypothesis as

$$H_0: p_1 = p_2$$

v/s

$H_1: P_1 \neq P_2$

Proportion z-test: $z = -3.4697094042513337$, $p = 0.0005210217431105547$

Here p value is less than 0.05 so we reject H_0 there is a statistically significant difference in the proportion.

students who passed between the two teaching methods are as proportion $P_{\text{online}} < P_{\text{offline}}$ that is Offline Method students Performed well.

Compare the proportion of students who passed (Passed = 1) between the two teaching methods (Online and Offline). Perform a two-sample z-test for proportions and interpret the findings.

5. Chi-square test of independence

H_0 : The two categorical variables are independent. v/s H_1 : The two categorical variables are dependent. There is association between Gender and Teaching Method. We Perform a chi-square test of independence using a contingency table of Gender $\times$ Method and get p-value = 0.5198 which is greater than so we **fail to reject H_0** that the two categorical variables **Gender , Method** are independent.

6. Test whether the teaching method has a significant effect on student Score.

Here we perform a one-way ANOVA and p-value is less than 0.05 so we reject H_0 that means there is a significance difference in teaching method . Now we proceed for post-hoc test or multiple comparison....

Then apply Tukey's HSD post-hoc test to identify pairwise differences. In Tukey's HSD post-hoc test for all group combination p-value less than 0.05 so all the teaching methods are making different effects on student(or all the methods are different).

7. Two-way ANOVA for effects of Method , Gender , and their interaction on Score .

Two-way ANOVA for effects of Method and Gender, have significant effects on Score because p-value less than 0.05. And interaction effects don't have significant effects on Score because p-value greater than 0.05.

8. Non-parametric alternative of Two sample independent t test (Mann-Whitney U test)

H_0 : $\text{Median}_{\text{online}}$ of Score = $\text{Median}_{\text{offline}}$ of Score v/s H_1 : $\text{Median}_{\text{online}}$ of Score $\neq$ $\text{Median}_{\text{offline}}$ of Score

When the assumption of normality is not met, we perform a Mann-Whitney U test to compare the median Score between the Online and Offline groups. And get. Mann-Whitney U = 112.5 , $p = 7.135872523782862e-09$ Here p value is less than 0.05 so we reject H_0 that Median are different .

9. Non-parametric alternative of Paired t-test (Wilcoxon signed-rank test)

As a non-parametric alternative to the paired t-test, perform a Wilcoxon signed-rank test to compare Score and After_Training.

H_0 : Score of Students before = Score of students after training v/s H_1 : Score of Students before $\neq$ Score of students after training

Wilcoxon W = 0.0 , $p = 2.177458322500681e-15$ Here p value is less than 0.05 so we reject H_0 that Student have got different Score after training.